Hotel Booking Cancellation Prediction

Decision Systems

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Data Dictionary

Variable	DESCRIPTION	BUSINESS RELEVANCE
BOOKING_ID	Unique identifier for each booking	Used to track individual bookings
NO_OF_ADULTS	The number of adults included in the booking	Helps estimate group size and potential revenue impact
NO_OF_CHILDREN	The number of children included in the booking	Similar to adults, influences revenue and service adjustments
NO_OF_WEEKEND_NIGHTS	Number of weekend nights booked or stayed	Can indicate premium time periods or special rates
NO_OF_WEEK_NIGHTS	Number of weekday nights booked or stayed	Helps in understanding weekday vs. weekend booking trends
REQUIRED_CAR_PARKING_SPACE	Indicator if a parking space is needed (0 = No, 1 = Yes)	May reflect customer preferences; sometimes tied to longer stays
ROOM_TYPE_RESERVED	Encoded type of room reserved by the customer	Can hint at customer segmentation (for ex: budget vs. premium guests)
LEAD_TIME	Number of days between the booking date and the arrival date	Longer lead times can correlate with higher uncertainty and cancellations
ARRIVAL_YEAR	Year of arrival	Useful for tracking trends over time
ARRIVAL_MONTH	Month of arrival	Identifies seasonality (peak vs. off-peak months)
ARRIVAL_DATE	The actual day of the month for arrival	Complements month information for finer trend analysis
MARKET_SEGMENT_TYPE	Segment category of the booking (for ex: online travel agency, direct booking, etc.)	Critical for distinguishing channels with higher cancellation risk
REPEATED_GUEST	Indicator if the customer is a returning guest (0 = No, 1 = Yes)	Returning guests are usually more loyal and less likely to cancel
NO_OF_PREVIOUS_CANCELLATIONS	Number of previous bookings canceled by the customer	Historical behavior is a strong predictor of future cancellations
NO_OF_PREVIOUS_BOOKINGS_NOT_CANCELED	Number of previous bookings where no cancellation occurred	Establishes reliability and booking consistency of the customer

Business Problem Overview

Problem:

- High cancellation rates are leading to revenue losses and operational challenges.
- Inconsistent booking patterns create forecasting uncertainties that affect staffing and resource allocation.

Solution Approach:

- Data Analysis: Examine historical booking and cancellation data using univariate and bivariate analyses to spot key trends.
- Predictive Modeling: Build interpretable models (Decision Trees and Random Forests) to assess cancellation risk.
- Business Interpretation: Translate technical insights into actionable strategies—adjust cancellation policies, optimize marketing channels, and improve customer retention.

EDA - Univariate Analysis

Strategies to Effectively Manage and Accommodate Increased Demand

• Forecasting & Planning:

- Use historical data and predictive insights to anticipate peak periods.

- Align staffing and resource allocation accordingly.

• Dynamic Pricing & Overbooking:

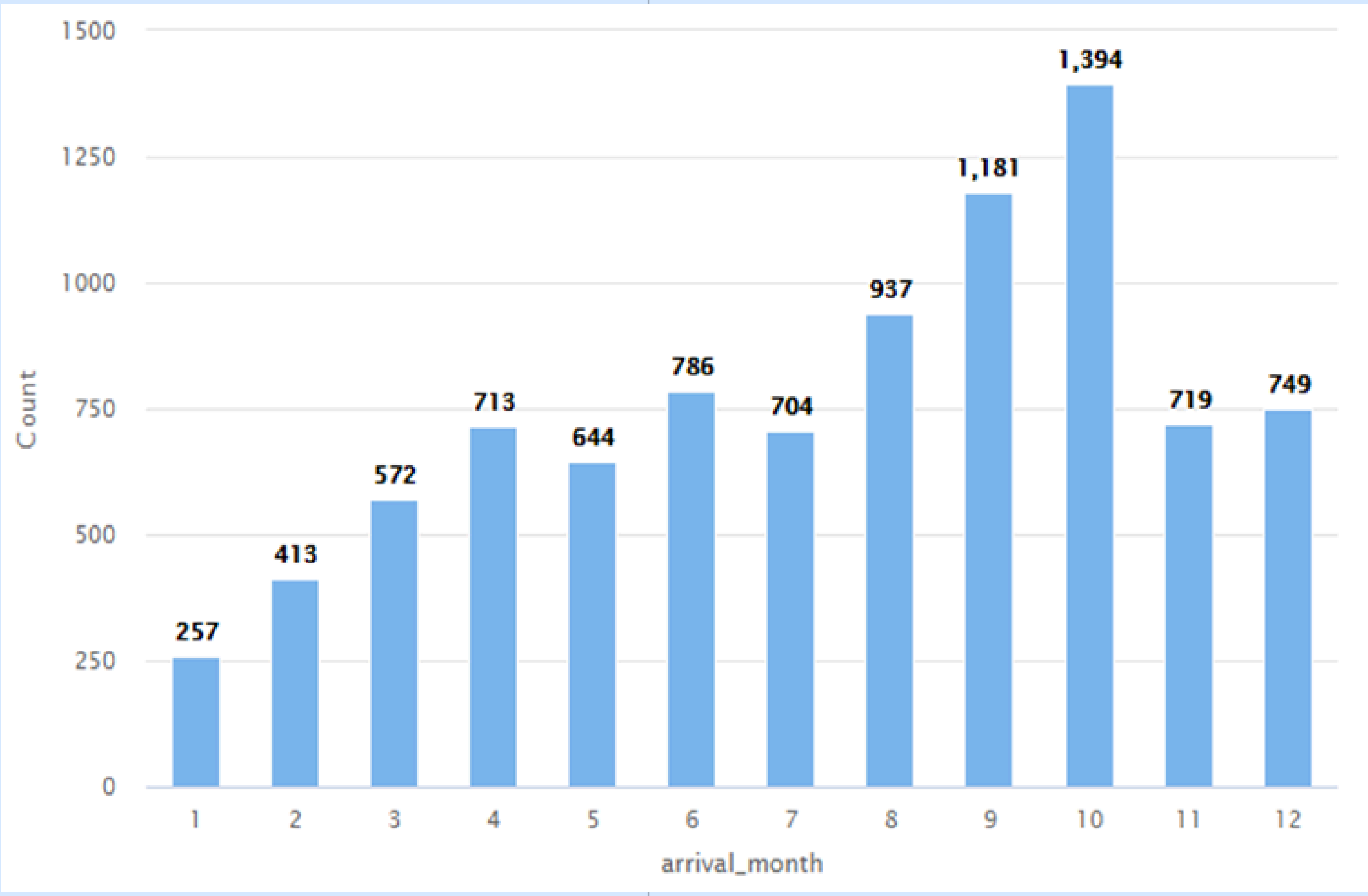
- Implement flexible pricing to optimize revenue during high-demand times.
- Consider controlled overbooking based on cancellation risk predictions.

• Enhanced Customer Engagement:

- Deploy targeted communication and personalized incentives to reduce last-minute cancellations.
- Strengthen loyalty programs to retain high-value customers.

• Operational Flexibility:

- Adapt operational processes to manage variability and ensure efficient service delivery during demand surges.

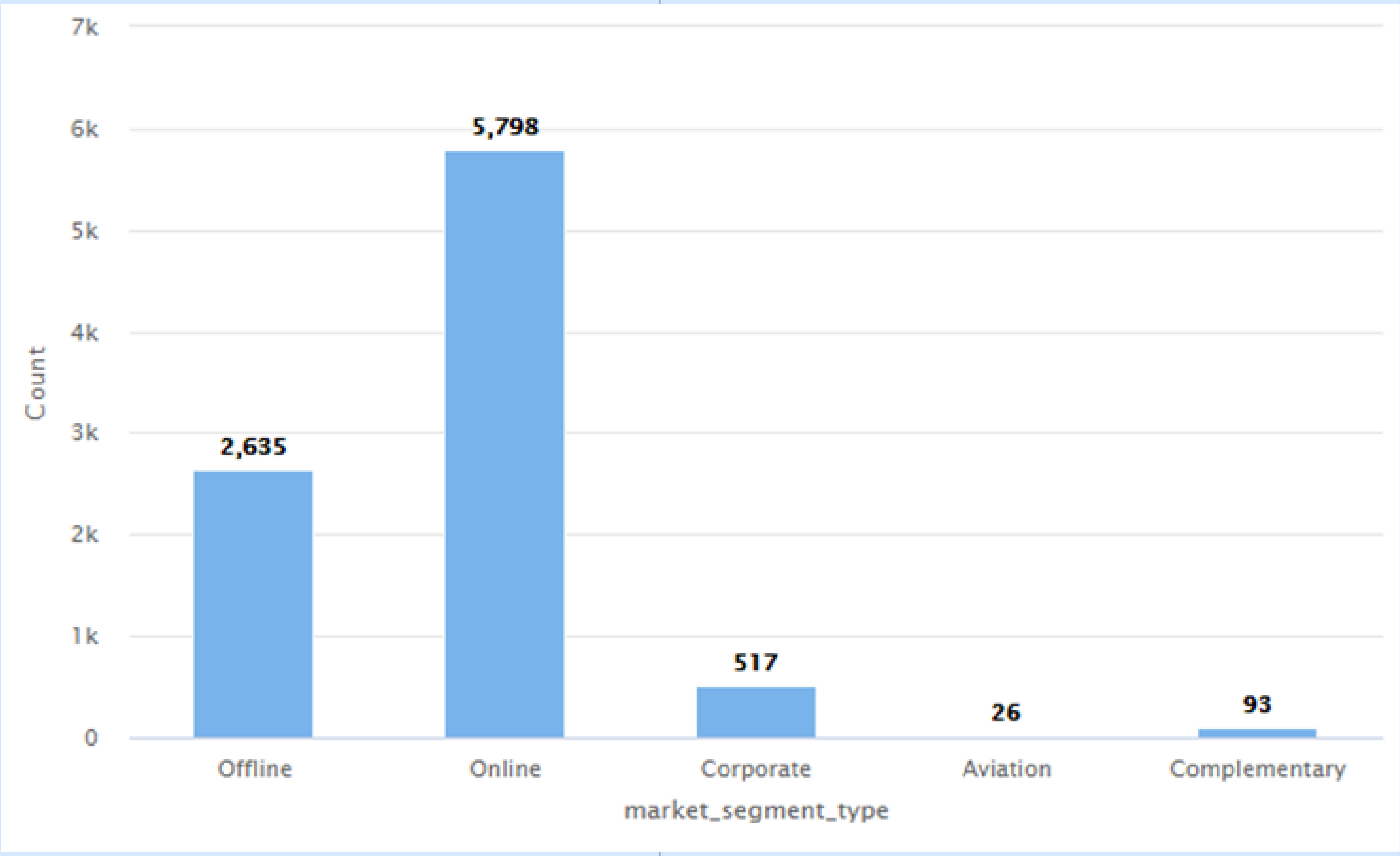


Online vs. Offline Bookings: Trends & Insights

Booking Channel Distribution:
- After analyzing the dataset, we found that 63.93% of bookings are made online, compared to 36.07% made offline.

• Key Insights:
- Digital Dominance: The majority of bookings are through online channels, reinforcing the importance of a strong digital presence.
- Opportunity for Offline Channels: The smaller offline share suggests potential for strategic enhancements (e.g., targeted promotions or improved in-person services) to boost their performance.

- Business Strategy: Tailor marketing and customer engagement to leverage the high volume of online bookings while also exploring ways to elevate the offline channel.



Cancellation Trends & Potential Solutions

Cancellation Analysis:

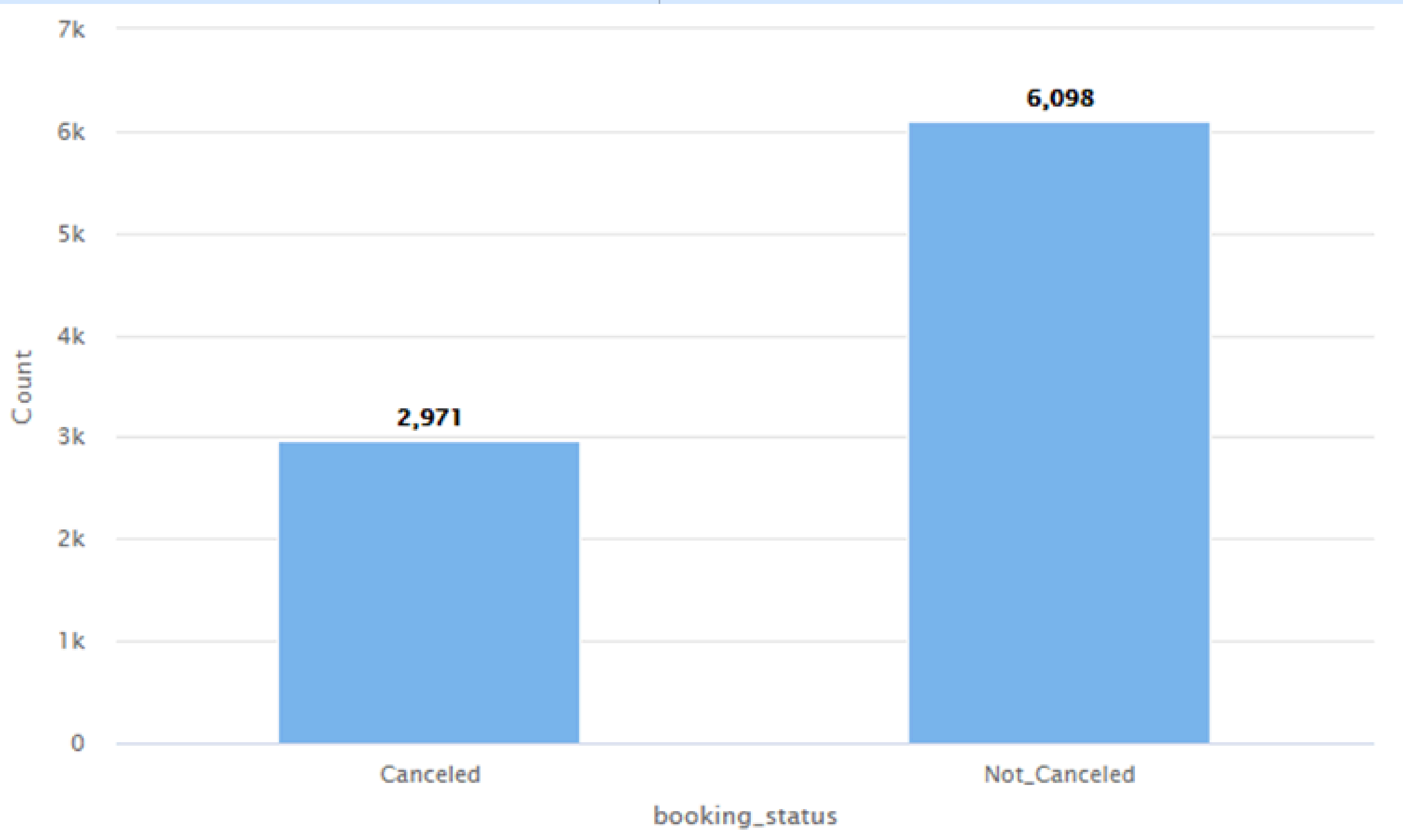
- Our analysis reveals that 2.76% of bookings are previously cancelled
- A cancellation rate in this range suggests notable revenue loss and operational disruption.

• **Is it a Serious Issue?**

- Yes. A high cancellation rate affects occupancy forecasting, staffing, and resource management while increasing revenue volatility.

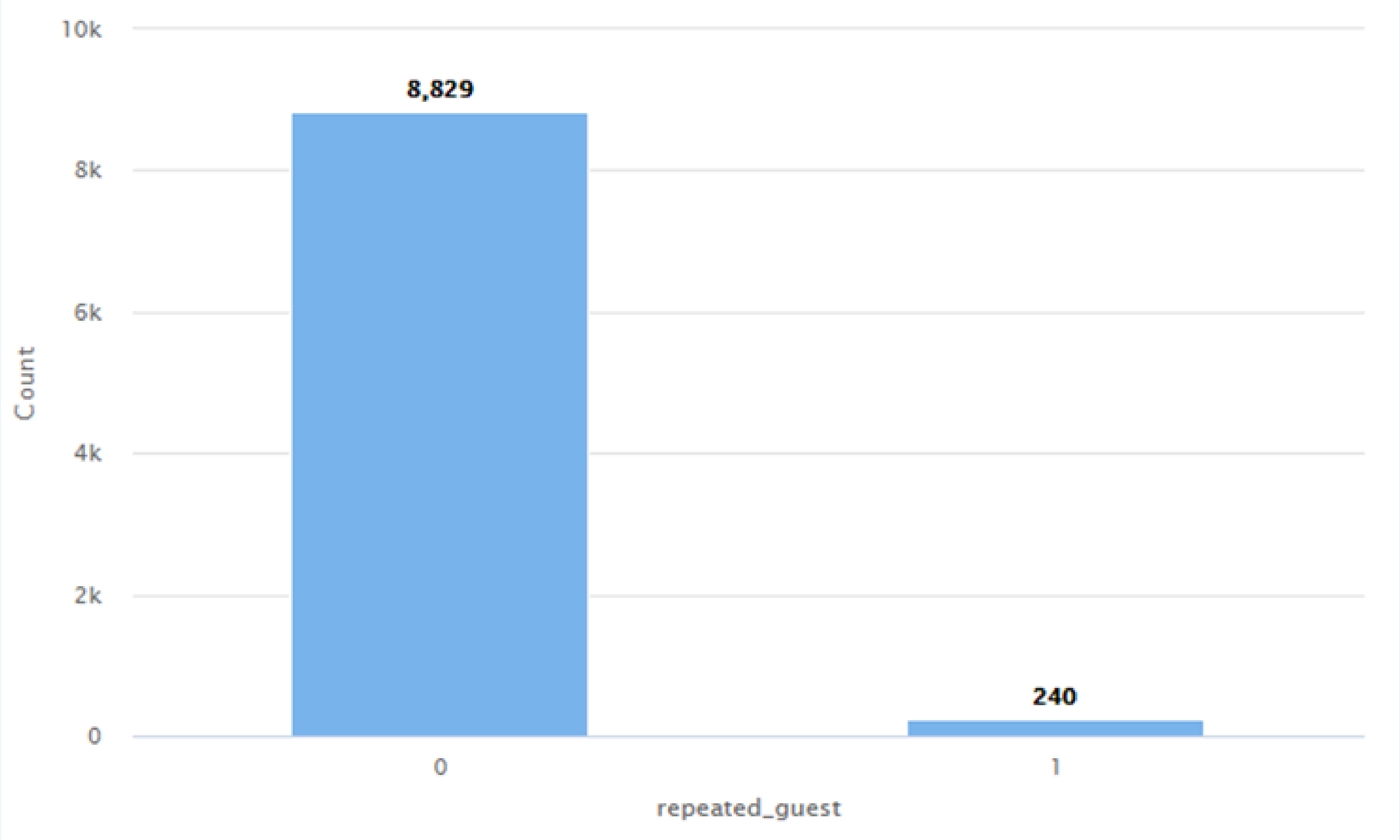
• **Potential Solutions:**

- **Revise Cancellation Policies:** Introduce or enforce non-refundable deposits for high-risk bookings or offer limited-time cancellation flexibility.
- **Enhanced Customer Communication:** Implement targeted reminders and confirmations to reduce last-minute cancellations.
- **Segmented Marketing:** Identify segments with higher cancellations and tailor incentives or loyalty programs to encourage commitment.
- **Predictive Analytics:** Use predictive models to flag high-risk bookings and proactively manage overbooking or customer engagement strategies.



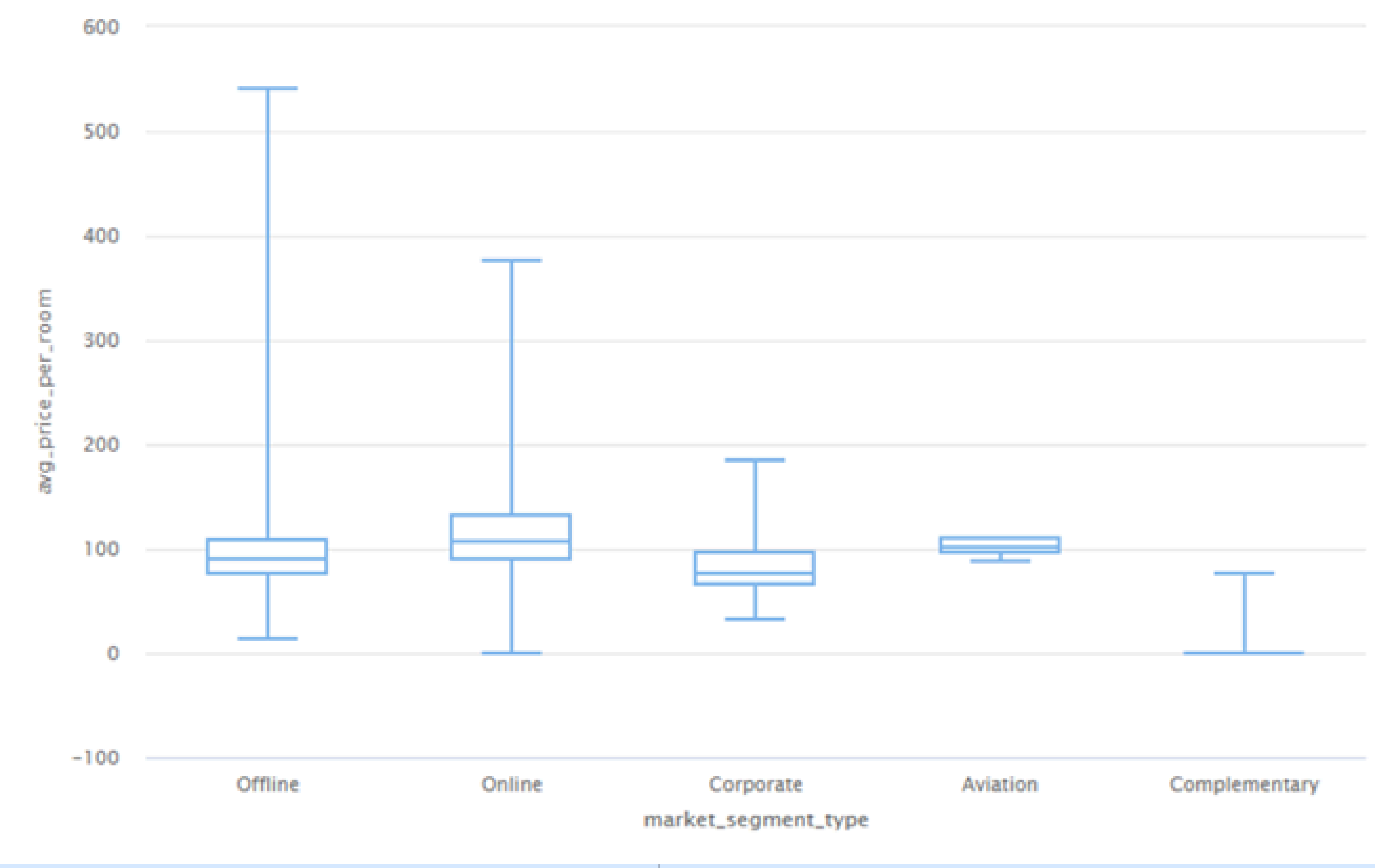
Repeated Guests Analysis

- **Observation:**
- Out of the total guest count, 2.4% are repeated visitors, indicating that a noticeable portion of the hotel's clientele returns more than once.
- **Positive or Negative Indication?**
- Positive: Repeated guests often reflect customer satisfaction and loyalty, driving a steady revenue stream.
- Negative: If the repeated guest percentage is low, it could imply that customer retention strategies need improvement.
- **Influencing Factors:**
- **Service Quality & Experience:** Memorable experiences encourage repeat stays.
- **Loyalty Programs & Incentives:** Rewarding regular visitors fosters brand preference.
- **Competitive Environment:** Guests might explore alternate accommodations if offers or locations are more appealing elsewhere.
- **Strategies for Improvement:**
- **Enhance Loyalty Programs:** Offer personalized perks (for ex: room upgrades, free meals).
- **Customer Feedback Loop:** Collect and act on guest feedback to continually improve service.
- **Targeted Marketing:** Use direct communication and special discounts for previous customers to encourage return visits.



Price Variations Across Market Segments: Revenue Management Insights

- **Strategic Implications:**
- **Segment-Specific Pricing:** Tailor room rates based on segment demand, willingness to pay, and booking patterns.
- **Promotional Offers:** Develop targeted promotions or discounts for specific segments to fill occupancy gaps or boost off-peak sales.
- **Yield Management:** Track real-time price elasticity in different segments to adjust rates dynamically and maximize revenue per available room.
- **Recommended Actions:**
- **Market Analysis:** Continuously monitor market trends and competitor pricing within each segment.
- **Cross-Selling & Upselling:** Offer relevant add-ons or packages (ex: spa, dining) aligned with each segment's preferences.
- **Data-Driven Adjustments:** Use predictive analytics to anticipate high-demand periods within each segment and optimize rates accordingly.

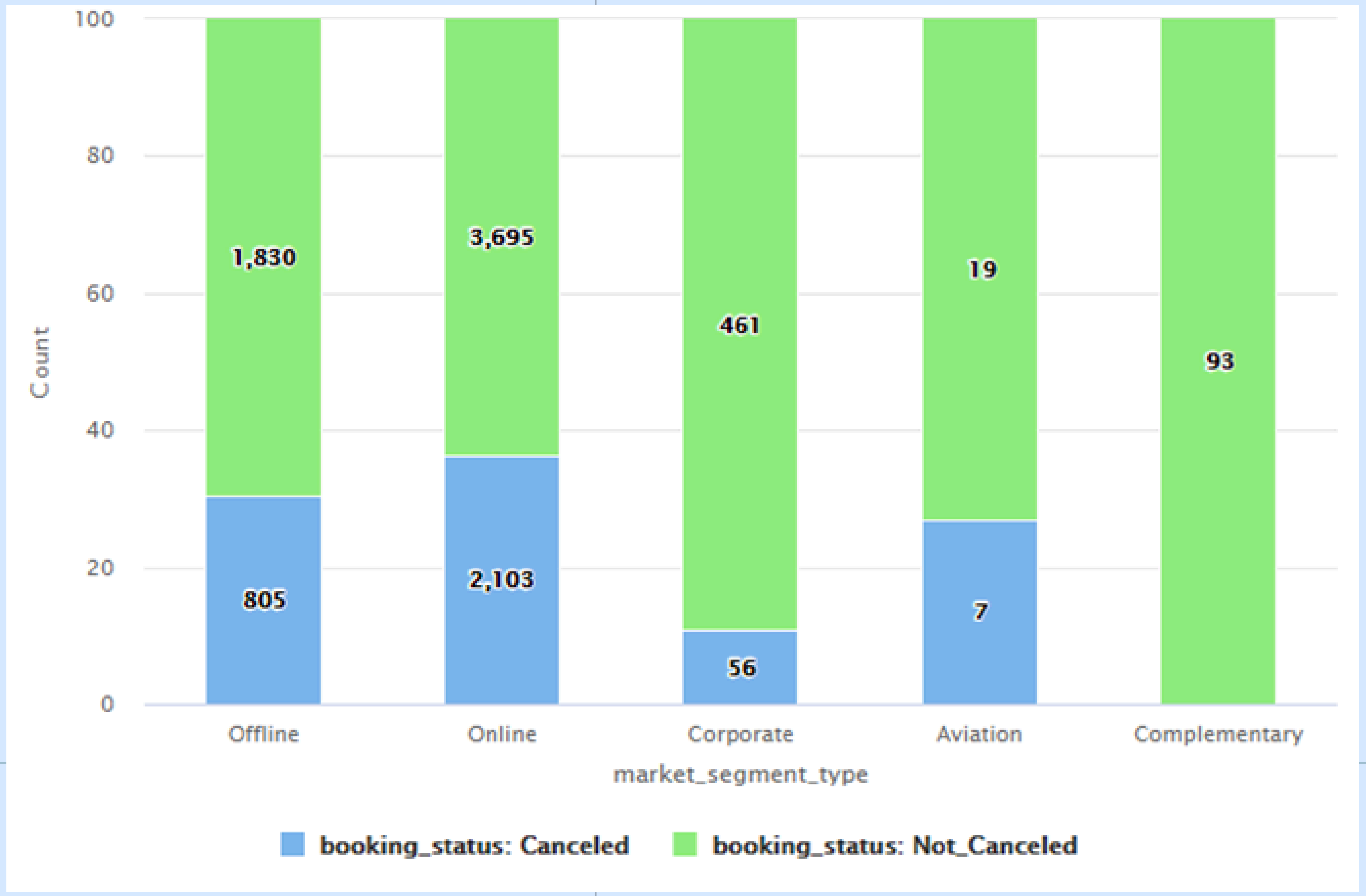


Overview of Univariate Analysis Discoveries

- **Seasonal Booking Trends:**
- Clear seasonal patterns were observed with peak booking volumes in specific months, suggesting opportunities to adjust promotions and staffing during high-demand periods.
- **Booking Channel Distribution:**
- The analysis shows a dominant trend of online bookings compared to offline. This underlines the importance of enhancing digital engagement and streamlining the online booking process.
- **Cancellation Rates:**
- A noticeable percentage of bookings are cancelled, which highlights an area of concern that needs addressing to mitigate revenue loss and improve forecast accuracy.
- **Repeated Guest Rates:**
- With only 2.4% of bookings coming from repeat guests, there is significant potential to strengthen customer loyalty programs and tailor retention strategies.
- **Price Variations Across Market Segments:**
- Distinct differences in average room prices among market segments suggest that differentiated pricing strategies could be implemented to optimize revenue management.

Cancellations Across Different Booking Channels: Insights & Mitigation Strategies

- **Observations:**
- The chart indicates varying cancellation rates across market segments (for ex: Online Travel Agencies, Direct Bookings, Corporate). Some segments exhibit higher cancellation percentages than others.
- **Key Insights:**
- Channel-Specific Risks: High-risk segments warrant stricter or more targeted cancellation policies, while lower-risk channels may allow more flexible terms.
- Customer Behavior Variances: Online channels often attract more price-sensitive and flexible travelers who may cancel if plans change or better deals appear.
- **Mitigation Strategies:**
- **Tailored Cancellation Policies:** For segments with higher cancellation rates, consider deposits or reduced cancellation windows.
- **Proactive Communication:** Send reminders and personalized offers to discourage last-minute cancellations, especially for online segment bookings.
- **Overbooking & Forecasting:** Use historical trends and predictive models to develop channel-specific overbooking strategies, optimizing occupancy.



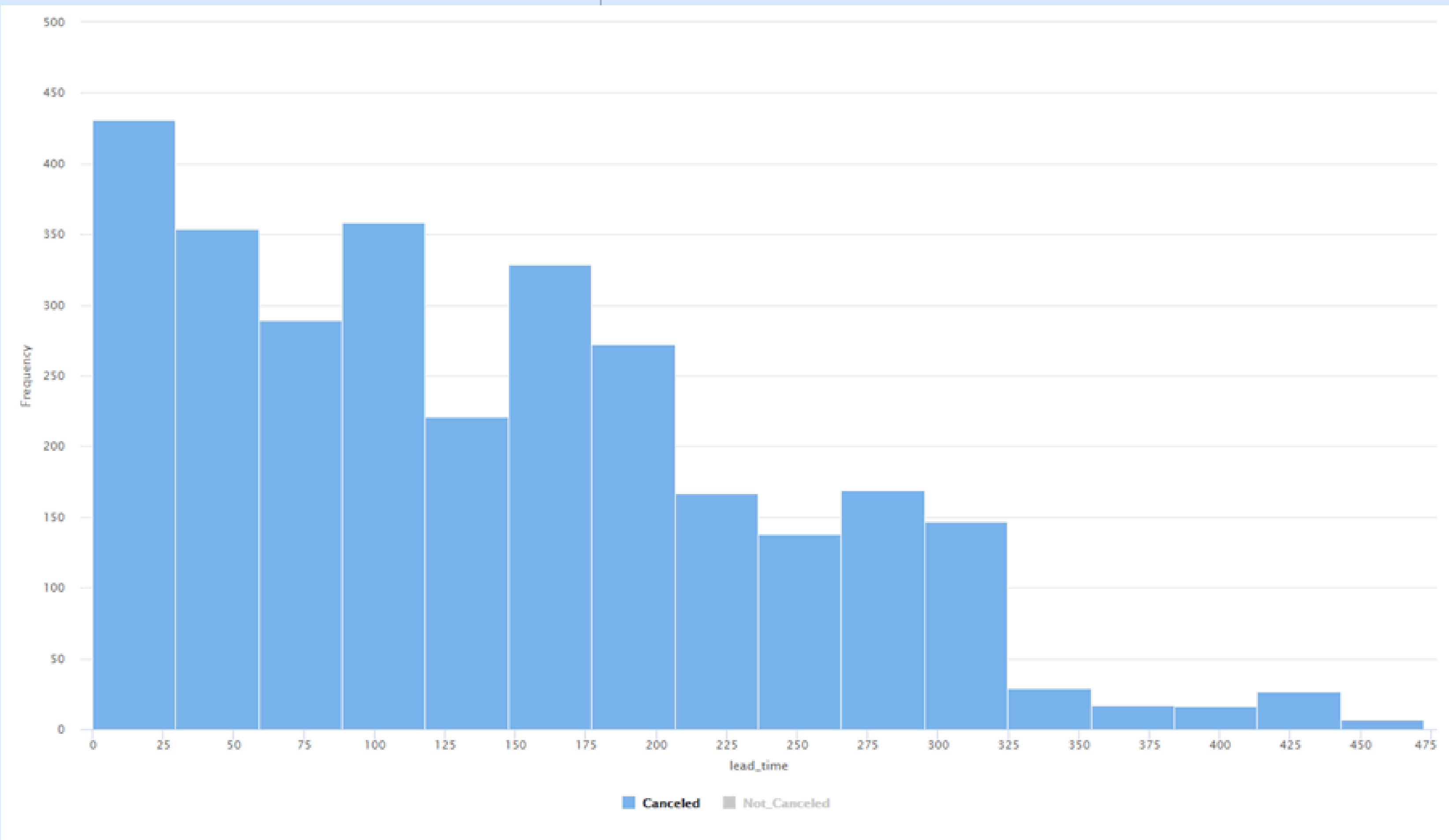
Special Requests vs. Booking Cancellations: Impact & Strategies

- **Observations:**
- The chart shows how bookings with special requests compare in cancellation rates to those without. A noticeable proportion of cancellations occurs within the segment that includes special requests.
- **Insights:**
- **Expectation Management:** Guests making special requests may have higher expectations, and unmet or misunderstood requests could lead to cancellations.
- **Operational Complexity:** Accommodating unique requests requires more coordination, potentially raising the chance of errors or dissatisfaction.
- **Mitigation Strategies:**
- **Enhanced Communication:** Proactively confirm and clarify special requests to manage guest expectations and reduce the chance of last-minute cancellations.
- **Priority Handling:** Allocate dedicated resources or a system to handle special requests efficiently, ensuring higher satisfaction rates.
- **Targeted Offers:** Provide incentives (for ex: upgrades, loyalty points) for guests who finalize special requests in advance, decreasing uncertainty and likelihood of cancellation.

Lead Time of Canceled Bookings: Insights & Opportunities

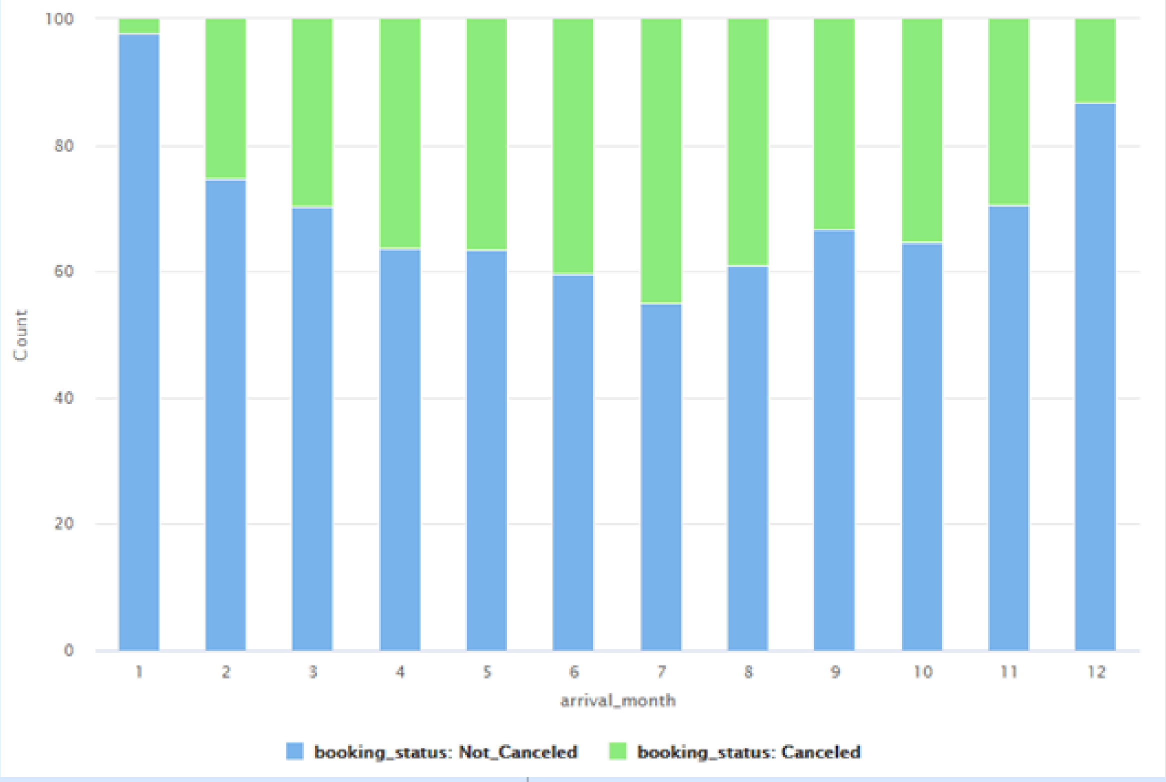
- **Observations:**
- This chart illustrates how far in advance customers book before ultimately canceling. A noticeable pattern emerges indicating that a significant portion of cancellations may occur when bookings are made far in advance (or at specific lead time ranges).
- **Insights:**
- **Long Lead Times → Higher Cancellation Likelihood:** Customers booking well ahead may have less certainty about their final travel plans, leading to increased cancellations.
- **Short-Notice Bookings:** These may either cancel less due to immediacy or are highly subject to last-minute changes—depending on the chart specifics.
- **Strategies to Improve Booking System & Customer Satisfaction:**

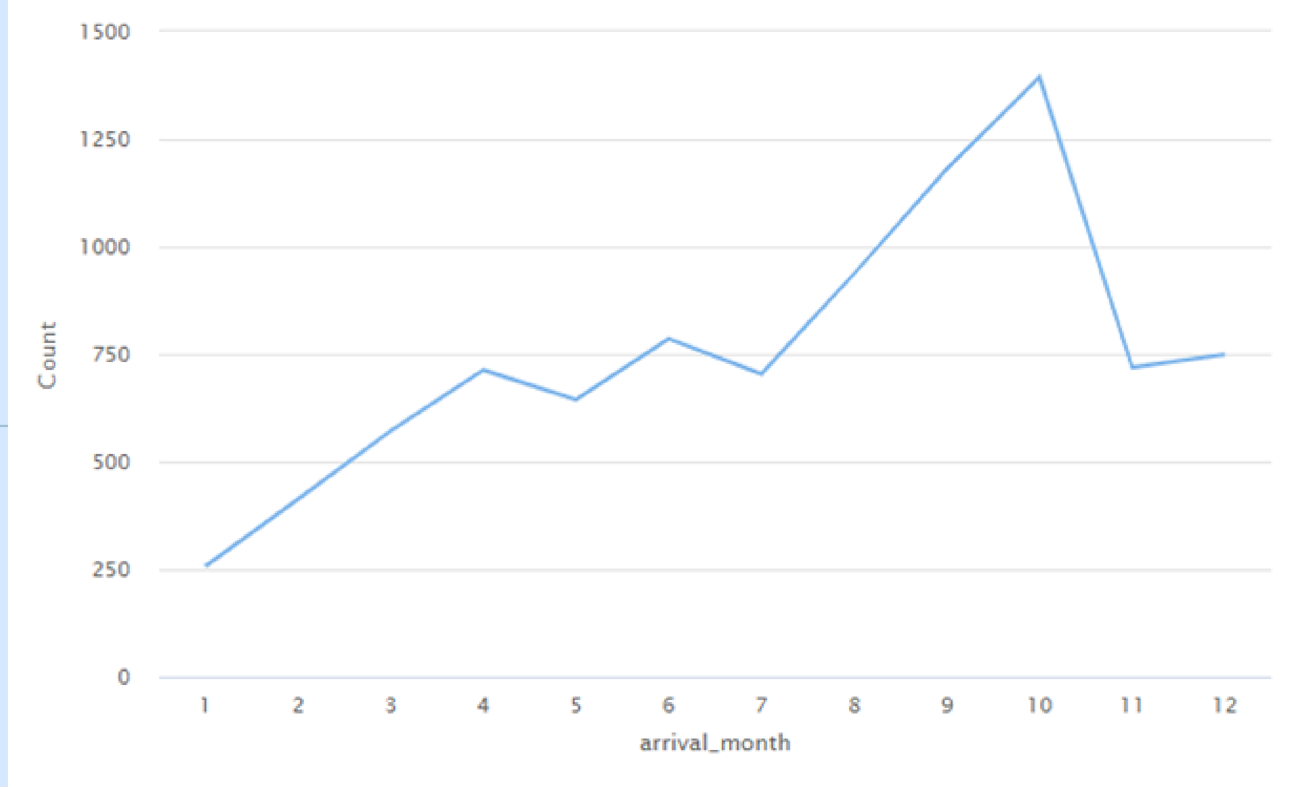
1. **Deposit & Incentive Policies:** For extended lead times, introduce partial or non-refundable deposits and/or offer early-bird discounts to secure commitment.
2. **Proactive Follow-Ups:** Send scheduled reminders or personalized offers to guests with longer lead times, encouraging them to confirm or modify plans well in advance.
3. **Flexible Booking Options:** Provide tiered cancellation policies (e.g., full refund until a certain period, then partial refunds) that still protect hotel revenue while appealing to customers' desire for flexibility.



Monthly Cancellation Rate Analysis: Patterns & Causes

- **Observations:**
- The line chart indicates fluctuating cancellation rates across different months. Peaks or dips may coincide with holiday seasons, local events, or business travel cycles.
- **Insights & Possible Reasons:**
- **Seasonal Variability:** High-demand periods can attract more speculative bookings, leading to higher cancellations.
- **External Influences:** Weather-related disruptions, economic factors, or major local events can cause abrupt shifts in cancellation patterns.
- **Travel Planning Habits:** Customers may cancel more frequently in certain months (for ex: shoulder seasons) if better deals or alternatives become available.
- **Actionable Strategies:**
- **Dynamic Pricing:** Adjust room rates based on historical trends and forecasted cancellation likelihood during peak vs. off-peak months.
- **Flexibility & Overbooking:** Implement slightly higher overbooking during months with historically elevated cancellations to maintain optimal occupancy.
- **Targeted Communication:** Use reminders or special promotions to reinforce bookings in months prone to higher cancellation rates, reducing last-minute drop-offs.





Overview of Bivariate Analysis Discoveries

- **Channel & Cancellation Relationship:**
- Cancellation rates vary by booking channel; certain segments (for ex: online channels) exhibit higher risk, highlighting the need for tailored policies.
- **Special Requests Impact:**
- Bookings with special requests tend to have elevated cancellation rates, indicating that managing guest expectations and service coordination is critical.
- **Lead Time Effects:**
- Longer lead times are generally associated with higher cancellation probabilities, suggesting that advanced bookings may benefit from stricter deposit or incentive policies.
- **Monthly Variations:**
- **Cancellation trends fluctuate monthly, influenced by seasonal demand and external factors, underscoring the importance of dynamic pricing and targeted communication strategies.**

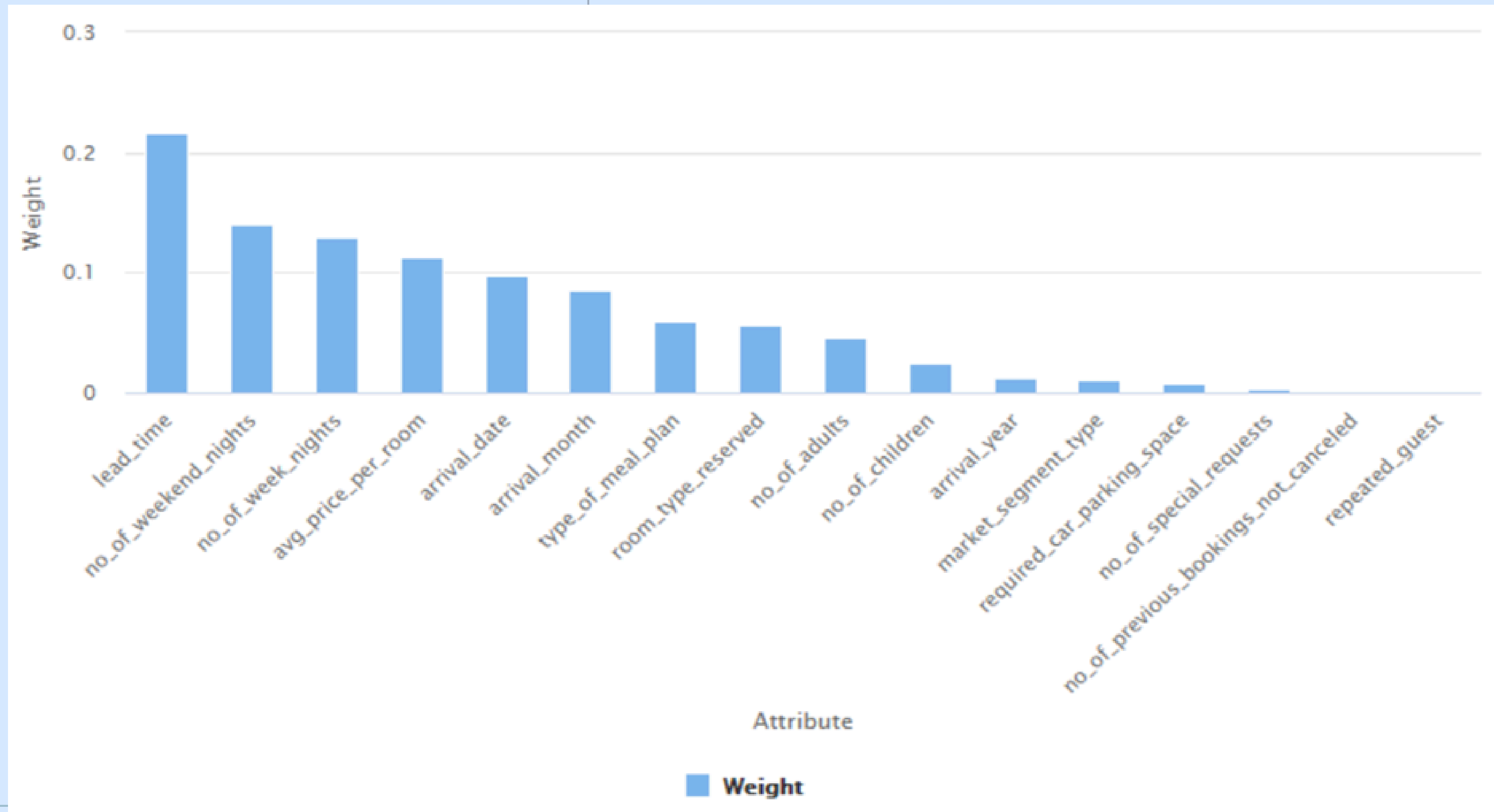
Model Performance Evaluation Decision Tree

- **Observations & Interpretations:**
- **Accuracy Drop:** The model achieves higher accuracy on the training data compared to the test data indicating some degree of overfitting.
- **Precision vs. Recall:**
- **Precision (76.27% on test):** Of all predicted cancellations, 76.27% were actually canceled. This is reasonably high, but not as strong as on the training set.
- **Recall (72.53% on test):** The model correctly identifies about 72.53% of actual cancellations on the test set. While decent, there is room for improvement in catching more true cancellations.
- **Overfitting Cues:** The gap between training and test metrics (especially accuracy and recall) suggests the model may be slightly tuned to training data patterns that don't generalize as well.

Model	Train Accuracy	Test Accuracy	Train Recall	Test Recall	Train Precision	Test Precision
Decision Tree	99.65	83.20	99.47	76.77	99.47	73.23

Model Performance Evaluation Decision Tree

- **High-Impact Features:**
- The top-ranked attributes (lead time, previous cancellations, deposit status) contribute most heavily to the Decision Tree's predictive power. The model relies on these features for its earliest splits, shaping how it separates likely cancellations from non-cancellations.
- **Lead Time Example:**
- If lead time is the most influential predictor, the model heavily factors in how far in advance a booking was made. Longer lead times may correlate with higher cancellation risk, and the Decision Tree uses this insight at key split points to boost accuracy.
- **Model Interpretability & Robustness:**
- Heavier weighting on a few critical features often improves interpretability by clearly illustrating the primary drivers behind cancellations. However, if these features are too dominant, the model may overlook other subtle factors, potentially contributing to overfitting.
- **Actionable Insights:**
- By focusing on the top-weighted attributes, hotel management can craft targeted interventions (for ex: deposit policies for long-lead bookings, incentives for repeat guests, etc.) that align with the model's strongest indicators of cancellation risk.



Model Performance Evaluation Pruned Decision Tree

- **Performance Comparison to Previous Version:**
- The pruned model shows a slight decrease in training metrics compared to the original tree but a noticeable improvement in test metrics (accuracy, precision, and recall).
- This indicates reduced overfitting and better generalization to unseen data, making the model more reliable for real-world predictions.
- **Factors Contributing to Improved Performance:**
- **Controlled Model Complexity:**
- Pruning removes less significant branches, preventing the tree from memorizing training data.
- This lowers variance and increases robustness across different data subsets.
- **Refined Decision Paths:**
- Shorter, more meaningful decision paths allow the model to focus on key variables (for ex: lead time, previous cancellations), improving interpretability while retaining predictive power.
- **Better Bias–Variance Trade-off:**
- By accepting a marginally higher bias (slightly lower training accuracy), the model significantly reduces variance, boosting test performance.
- **Business Implications:**
- **More Reliable Predictions:** The pruned tree provides a more consistent identification of high-risk bookings, enabling proactive cancellation management.
- **Actionable Insights:** Clearer, simpler rules facilitate stakeholder understanding and support targeted interventions (for ex: deposit policies, loyalty strategies) without overwhelming complexity.

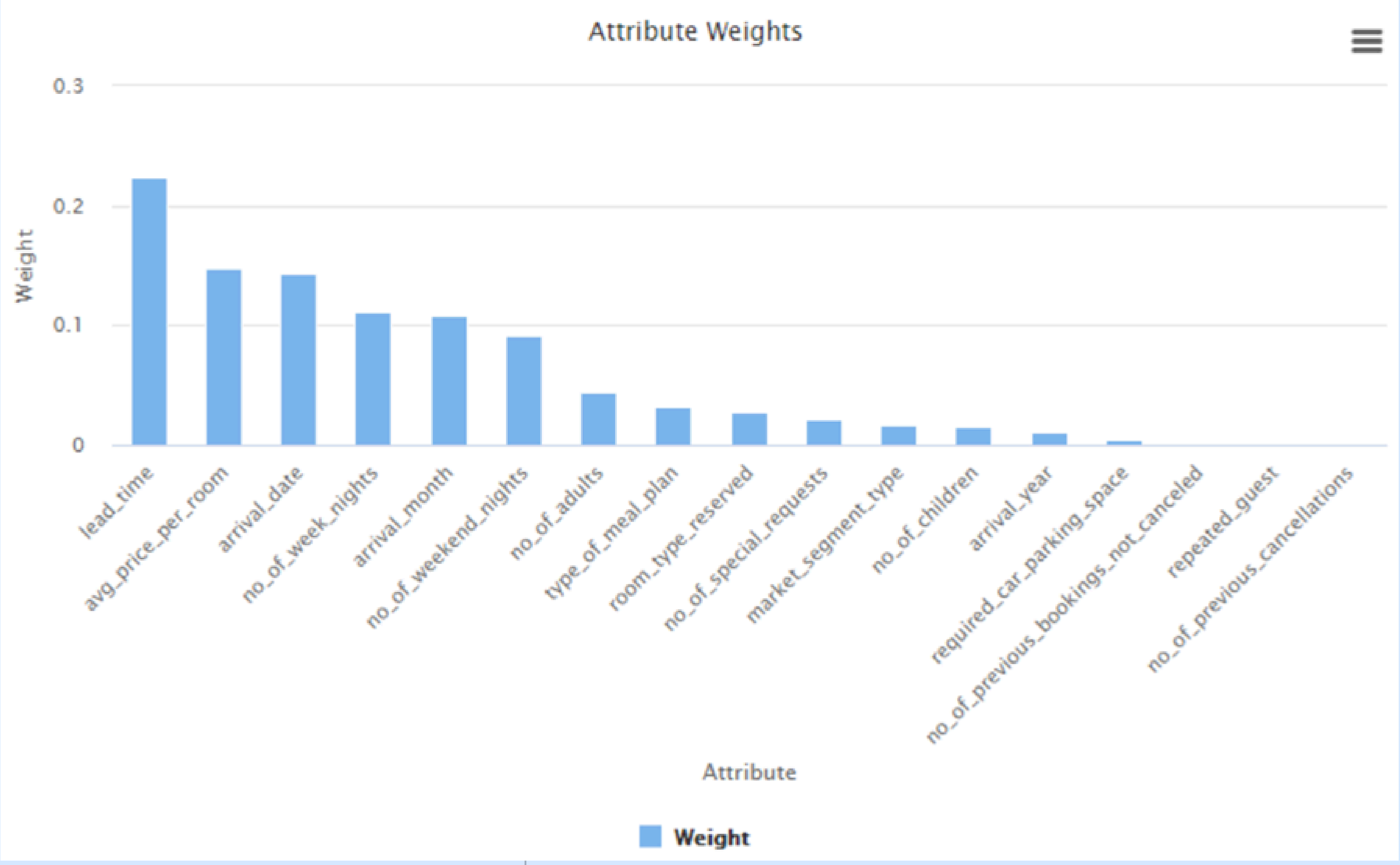
Model	Train Accuracy	Test Accuracy	Train Recall	Test Recall	Train Precision	Test Precision
Decision Tree	99.65	83.20	99.47	76.77	99.47	73.23
Decision Tree - Pruned	95.64	84.34	90.77	76.09	95.69	76.09

Model Performance Evaluation Random Forest

- **Top Contributing Features:**
- Features such as lead time, previous cancellations, and deposit status typically rank highest in feature importance for predicting cancellation risk. Their high weights indicate a strong influence on the Random Forest's splitting logic.
- **Influence on Overall Performance:**

1. **Model Focus:**
    - By concentrating on the most critical predictors, the Random Forest efficiently separates cancellations from non-cancellations, driving strong accuracy and precision.
2. **Bias vs. Variance:**
    - Overemphasis on a few features could mean the model misses subtle patterns from lower-weight features, potentially leading to reduced recall or slight overfitting.
3. **Interpretability & Actionability:**
    - Knowing the top attributes (e.g., lead time) helps management prioritize targeted policies—such as deposit requirements for long lead times—directly aligning data-driven insights with operational decisions.

- **Business Impact:**
- Identifying and focusing on these high-weight attributes allows hotels to develop segment-specific interventions, optimize cancellation policies, and refine marketing strategies. This ensures more precise risk management and proactive decision-making to reduce potential revenue losses.



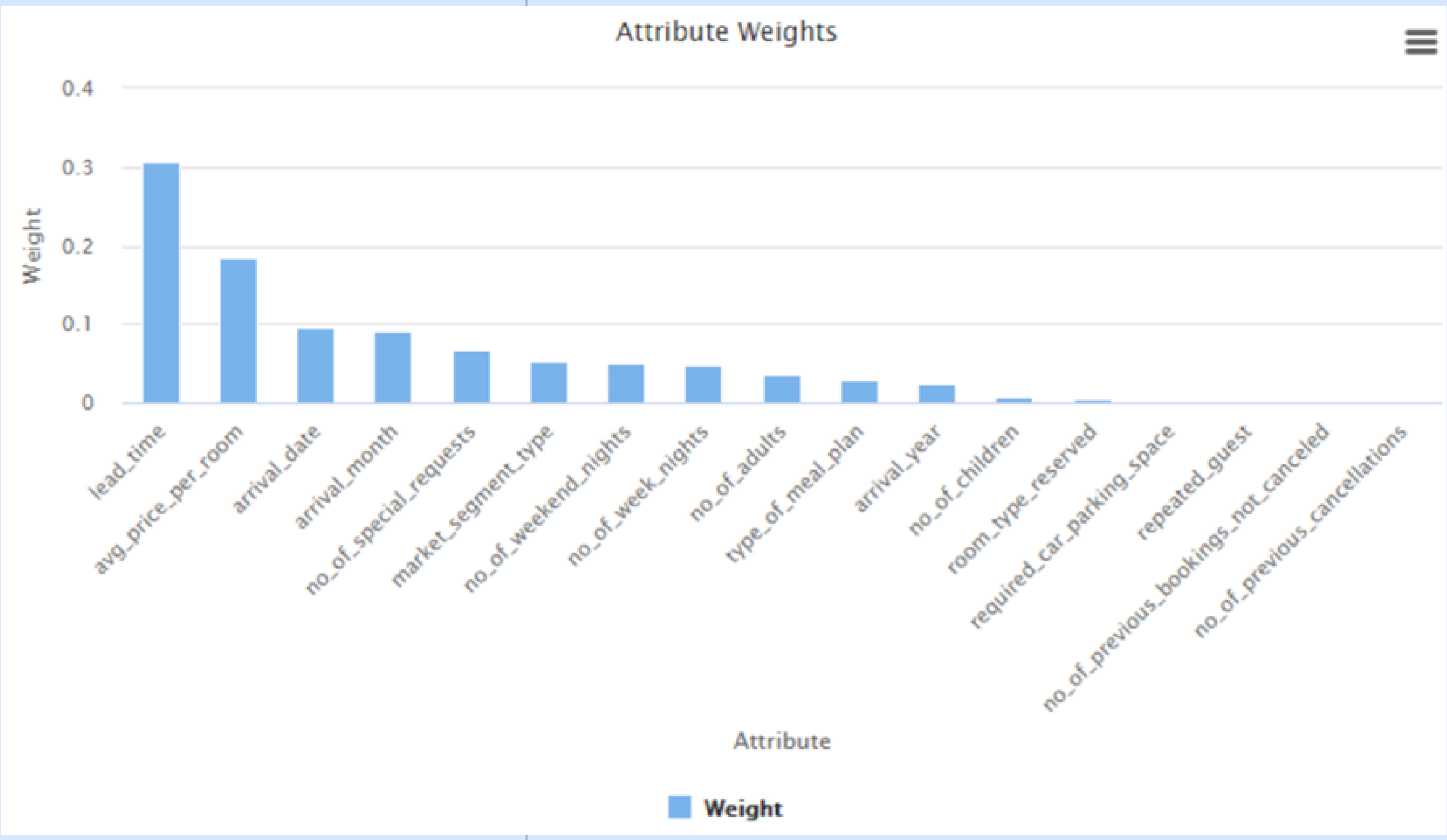
Model Performance Evaluation
Pruned Random Forest

- **Comparison to the Original Model:**
- The pruned Random Forest shows a smaller gap between training and test performance, indicating reduced overfitting.
- Although there may be a slight decline in training metrics, the test metrics (accuracy, precision, and recall) generally improve or remain more consistent compared to the unpruned version.
- **Factors Behind Improved Performance:**
- **Reduced Model Complexity:**
- Pruning eliminates less significant trees or branches, simplifying the ensemble. This helps the model generalize better and avoid memorizing spurious patterns in the training data.
- **Enhanced Generalization:**
- With fewer unnecessary splits, the pruned Random Forest focuses on the most impactful features, leading to more reliable predictions on unseen data.
- **Better Bias–Variance Trade-Off:**
- Sacrificing a small amount of training accuracy (slightly higher bias) yields a notable improvement in test accuracy and recall (lower variance).
- **Implications for Business:**
- More Robust Predictions: A pruned model that better generalizes is more trustworthy for operational decision-making, such as identifying high-risk bookings.
- **Targeted Interventions:** By highlighting key predictive features, the pruned Random Forest helps management focus on lead time, deposit policies, or marketing campaigns for risk reduction.
- **Optimized Resource Allocation:** With improved out-of-sample performance, overbooking strategies and revenue projections can be more precisely tuned, minimizing lost opportunities and costs.

Model	Train Accuracy	Test Accuracy	Train Recall	Test Recall	Train Precision	Test Precision
Random Forest	98.47	87.94	96.44	77.10	98.87	84.71
Random Forest - Pruned	87.32	85.04	69.86	66.22	89.09	84.77

Model Performance Evaluation
Pruned Random Forest

- **Dominant Features:**
- Similar to the unpruned model, features such as lead time, previous cancellations, and deposit status typically rank highest. Pruning ensures the model focuses on these key variables, strengthening predictive accuracy without overfitting.
- **Influence on Overall Performance:**
- **Streamlined Feature Utilization:**
- Pruning reduces the number of inconsequential splits, giving more weight to critical attributes. This yields clearer, more powerful decision pathways.
- **Balanced Complexity:**
- By limiting the tree depths and cutting back minor splits, the pruned Random Forest achieves a solid bias–variance trade-off, improving test performance and generalization.
- **Actionable Insights:**
- When pivotal features (for ex: lead time) emerge as top drivers of cancellation, hotel managers can prioritize strategies around early-booker deposits or tailored follow-ups.
- **Business Implications:**
- **Targeted Interventions:** Focusing on high-weight attributes drives more effective cancellation mitigation, such as requiring a deposit for longer lead times.
- **Improved Decision Support:** A pruned Random Forest with a clear feature hierarchy offers easily interpretable rules, making it simpler for business stakeholders to act on data-driven insights.



Model Performance Summary

- **Best-Fit Model for Business Objective:**
- The Pruned Random Forest strikes the best balance between accuracy, precision, and recall, making it ideal for identifying high-risk cancellations. Its improved generalization minimizes false negatives (missed cancellations), which is crucial for proactive revenue management.
- **Important Evaluation Metric:**
- Recall is particularly important in this use case, as the hotel aims to capture as many potential cancellations as possible. A higher recall ensures the hotel can take timely action (for ex: deposits, reminders) to reduce actual cancellations.
- **Key Predictive Features:**
- **Lead Time:** Longer lead times often correlate with higher cancellation risk.
- **Previous Cancellations:** Guests with a history of canceling are more prone to repeat cancellations.
- **Deposit Status:** Prepaid or partially paid bookings are less likely to cancel.
- **Suggestions for the Hotel:**
- **Refine Cancellation Policies:**
- Consider deposits or stricter terms for long lead-time bookings to discourage speculative reservations.
- **Targeted Follow-Ups:**
- Send reminders or personalized incentives to guests flagged with a higher risk of cancellation.
- **Dynamic Pricing & Overbooking:**
- Adjust room rates and manage slight overbookings, using the model's predictions to maintain optimal occupancy.
- **Loyalty & Engagement:**
- Encourage returning guests through tailored loyalty programs, focusing on those who have previously canceled but still show willingness to book again.
- **By incorporating the Pruned Random Forest predictions and insights into everyday operations, the hotel can more effectively mitigate cancellations, improve revenue stability, and enhance overall guest satisfaction.**

Model	Train Accuracy	Test Accuracy	Train Recall	Test Recall	Train Precision	Test Precision
Decision Tree	99.65	83.20	99.47	76.77	99.47	73.23
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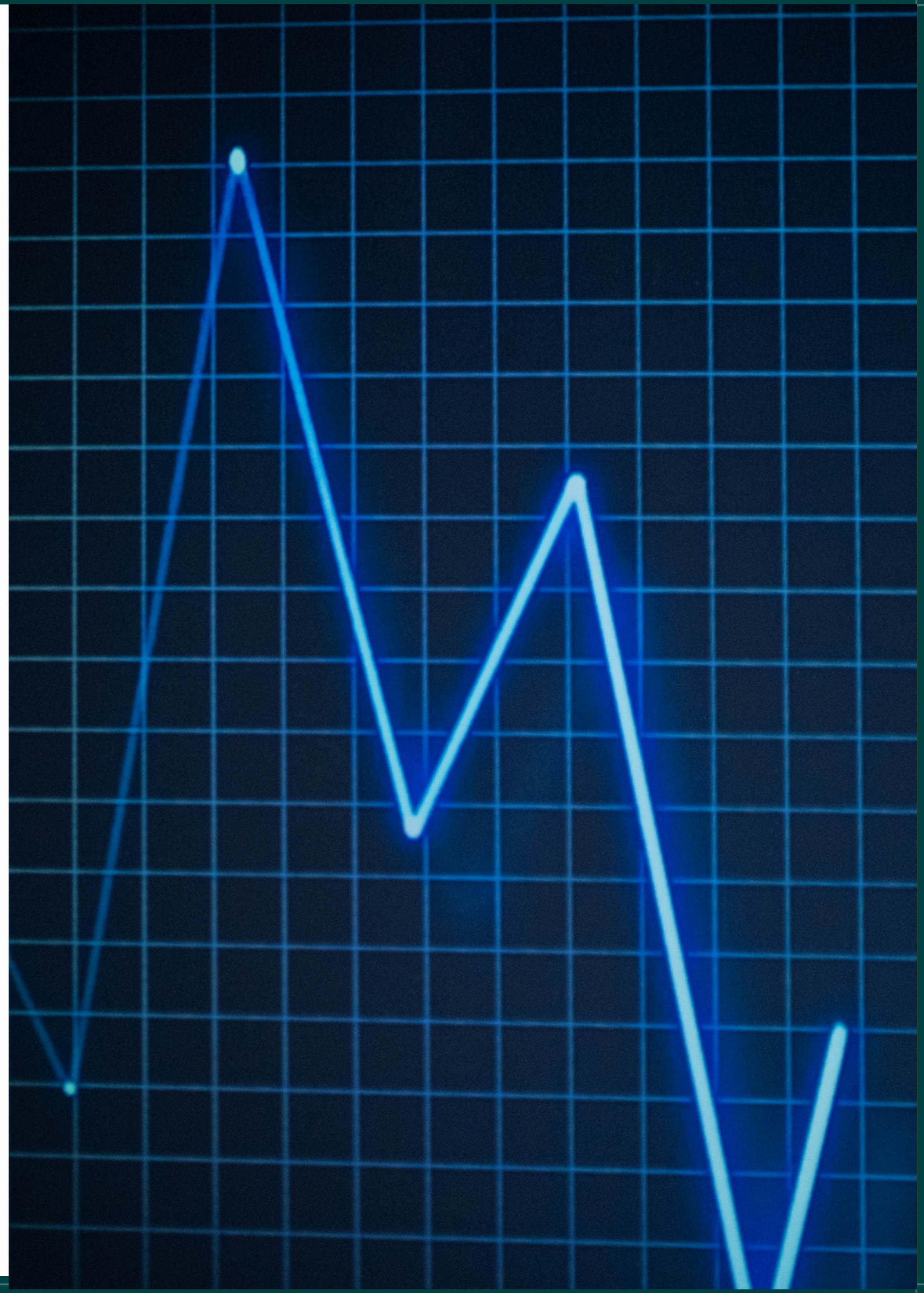

Conclusion

Our analysis revealed that long lead times, previous cancellations, and deposit policies are major drivers of booking cancellations. By leveraging a Pruned Random Forest model, we gained actionable insights that can help the hotel identify high-risk bookings and adjust policies accordingly. Implementing targeted interventions such as refined cancellation policies, enhanced customer engagement, and dynamic pricing strategies can significantly mitigate cancellation risks, ultimately optimizing revenue and improving guest satisfaction.